



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Solve a 2×2 system by substitution
- Solve a 2×2 system by elimination
- Interpret the solution of a system in context

Simplify each expression completely. Show all steps and circle your final answer.

Solve a 2×2 linear system

1. Solve the system: $3x + 1y = -13$ and $2x + 1y = -10$.

$$3x + 1y = -13, 2x + 1y = -10$$

Answer: _____

2. Solve the system by substitution: $y = 3x + -2$ and $4x + 2y = -12$.

$$y = 3x + -2, 4x + 2y = -12$$

Answer: _____

3. Solve the system by elimination: $2x - 2y = -2$ and $2x + 1y = 13$.

$$2x - 2y = -2, 2x + 1y = 13$$

Answer: _____

4. Solve the system: $4x + 1y = 2$ and $3x + 2y = 4$.

$$4x + 1y = 2, 3x + 2y = 4$$

Answer: _____

5. Solve the system by substitution: $y = 1x + -1$ and $2x + 3y = 0$.

$$y = 1x + -1, 2x + 3y = 0$$

Answer: _____

6. Solve the system by elimination: $5x - 3y = 12$ and $4x + 2y = 14$.

$$5x - 3y = 12, 4x + 2y = 14$$

Answer: _____

7. Solve the system: $2x + 3y = 11$ and $3x + 1y = 13$.

$$2x + 3y = 11, 3x + 1y = 13$$

Answer: _____

8. Solve the system by substitution: $y = 2x + -3$ and $3x + 3y = -6$.

$$y = 2x + -3, 3x + 3y = -6$$

Answer: _____

9. Solve the system by elimination: $4x - 3y = 7$ and $2x + 2y = 14$.

$$4x - 3y = 7, 2x + 2y = 14$$

Answer: _____

10. Solve the system: $2x + 1y = 2$ and $2x + 3y = 10$.

$$2x + 1y = 2, 2x + 3y = 10$$

Answer: _____

11. Solve the system by substitution: $y = 3x + 4$ and $4x + 2y = -12$.

$$y = 3x + 4, 4x + 2y = -12$$

Answer: _____

12. Solve the system by elimination: $5x - 3y = 19$ and $4x + 3y = 26$.

$$5x - 3y = 19, 4x + 3y = 26$$

Answer: _____

13. Solve the system: $3x + 2y = 0$ and $4x + 2y = -2$.

$$3x + 2y = 0, 4x + 2y = -2$$

Answer: _____

14. Solve the system by substitution: $y = 1x + 0$ and $1x + 3y = -6$.

$$y = 1x + 0, 1x + 3y = -6$$

Answer: _____

15. Solve the system by elimination: $5x - 3y = 22$ and $3x + 2y = 17$.

$$5x - 3y = 22, 3x + 2y = 17$$

Answer: _____

16. Solve the system: $1x + 3y = -1$ and $2x + 2y = -6$.

$$1x + 3y = -1, 2x + 2y = -6$$

Answer: _____

17. Solve the system by substitution: $y = 1x + -1$ and $1x + 1y = 0$.

$$y = 1x + -1, 1x + 1y = 0$$

Answer: _____

18. Solve the system by elimination: $5x - 1y = 9$ and $1x + 2y = 4$.

$$5x - 1y = 9, 1x + 2y = 4$$

Answer: _____

19. Solve the system: $2x + 1y = 0$ and $3x + 2y = 0$.

$$2x + 1y = 0, 3x + 2y = 0$$

Answer: _____

20. Solve the system by substitution: $y = 3x + -1$ and $3x + 2y = -4$.

$$y = 3x + -1, 3x + 2y = -4$$

Answer: _____

21. Solve the system by elimination: $2x - 2y = 6$ and $2x + 2y = 14$.

$$2x - 2y = 6, 2x + 2y = 14$$

Answer: _____

22. Solve the system: $3x + 2y = 15$ and $3x + 3y = 18$.

$$3x + 2y = 15, 3x + 3y = 18$$

Answer: _____

23. Solve the system by substitution: $y = 3x + 2$ and $1x + 1y = -3$.

$$y = 3x + 2, 1x + 1y = -3$$

Answer: _____

24. Solve the system by elimination: $4x - 2y = 10$ and $2x + 2y = 14$.

$$4x - 2y = 10, 2x + 2y = 14$$

Answer: _____

25. Solve the system: $4x + 2y = -12$ and $2x + 2y = -6$.

$$4x + 2y = -12, 2x + 2y = -6$$

Answer: _____

26. Solve the system by substitution: $y = 2x + 3$ and $4x + 2y = -18$.

$$y = 2x + 3, 4x + 2y = -18$$

Answer: _____

27. Solve the system by elimination: $2x - 2y = -2$ and $2x + 2y = 18$.

$$2x - 2y = -2, 2x + 2y = 18$$

Answer: _____

28. Solve the system: $2x + 3y = -17$ and $1x + 2y = -10$.

$$2x + 3y = -17, 1x + 2y = -10$$

Answer: _____

29. Solve the system by substitution: $y = 3x + 3$ and $4x + 1y = -12$.

$$y = 3x + 3, 4x + 1y = -12$$

Answer: _____

30. Solve the system by elimination: $2x - 1y = 1$ and $3x + 1y = 14$.

$$2x - 1y = 1, 3x + 1y = 14$$

Answer: _____



MATH130: Systems of Equations

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ANSWER KEY & SOLUTIONS

Topics: Solve a 2x2 linear system. All answers verified by independent computation.

Solutions

Solve a 2x2 linear system

1. Solve the system: $3x + 1y = -13$ and $2x + 1y = -10$.

$$3x + 1y = -13, 2x + 1y = -10$$

→ From the coefficients and solution, $c = 3(-3) + 1(-4) = -13$.

→ $f = 2(-3) + 1(-4) = -10$.

→ Using elimination: multiply Eq1 by 2 and Eq2 by 3, subtract to solve for y , then back-substitute.

→ Solution: $x = -3$, $y = -4$.

Answer: $x = -3$, $y = -4$

2. Solve the system by substitution: $y = 3x + -2$ and $4x + 2y = -12$.

$$y = 3x + -2, 4x + 2y = -12$$

→ Substitute $y = 3x + -2$ into $4x + 2y = -12$.

→ $4x + 2(3x + -2) = -12 \Rightarrow (4 + 6)x = -12 - -4 \Rightarrow x = -3$.

→ $y = 3(-3) + -2 = 0$.

Answer: $x = -3$, $y = 0$

3. Solve the system by elimination: $2x - 2y = -2$ and $2x + 1y = 13$.

$$2x - 2y = -2, 2x + 1y = 13$$

→ Add the equations to eliminate y : $(2+2)x = -2 + 13 \Rightarrow 4x = 11$.

→ $x = 4$. Substitute back to find $y = 5$.

Answer: $x = 4$, $y = 5$

4. Solve the system: $4x + 1y = 2$ and $3x + 2y = 4$.

$$4x + 1y = 2, 3x + 2y = 4$$

→ From the coefficients and solution, $c = 4(0) + 1(2) = 2$.

→ $f = 3(0) + 2(2) = 4$.

→ Using elimination: multiply Eq1 by 3 and Eq2 by 4, subtract to solve for y , then back-substitute.

→ Solution: $x = 0$, $y = 2$.

Answer: $x = 0$, $y = 2$

5. Solve the system by substitution: $y = 1x + -1$ and $2x + 3y = 0$.

$$y = 1x + -1, 2x + 3y = 0$$

→ Substitute $y = 1x + -1$ into $2x + 3y = 0$.

→ $2x + 3(1x + -1) = 0 \Rightarrow (2 + 3)x = 0 - -3 \Rightarrow x = -3$.

→ $y = 1(-3) + -1 = 2$.

Answer: $x = -3$, $y = 2$

6. Solve the system by elimination: $5x - 3y = 12$ and $4x + 2y = 14$.

$$5x - 3y = 12, 4x + 2y = 14$$

→ Add the equations to eliminate y : $(5+4)x = 12 + 14 \Rightarrow 9x = 26$.

→ $x = 3$. Substitute back to find $y = 1$.

Answer: $x = 3, y = 1$

7. Solve the system: $2x + 3y = 11$ and $3x + 1y = 13$.

$$2x + 3y = 11, 3x + 1y = 13$$

→ From the coefficients and solution, $c = 2(4) + 3(1) = 11$.

→ $f = 3(4) + 1(1) = 13$.

→ Using elimination: multiply Eq1 by 3 and Eq2 by 2, subtract to solve for y , then back-substitute.

→ Solution: $x = 4, y = 1$.

Answer: $x = 4, y = 1$

8. Solve the system by substitution: $y = 2x + -3$ and $3x + 3y = -6$.

$$y = 2x + -3, 3x + 3y = -6$$

→ Substitute $y = 2x + -3$ into $3x + 3y = -6$.

→ $3x + 3(2x + -3) = -6 \Rightarrow (3 + 6)x = -6 - -9 \Rightarrow x = -3$.

→ $y = 2(-3) + -3 = 1$.

Answer: $x = -3, y = 1$

9. Solve the system by elimination: $4x - 3y = 7$ and $2x + 2y = 14$.

$$4x - 3y = 7, 2x + 2y = 14$$

→ Add the equations to eliminate y : $(4+2)x = 7 + 14 \Rightarrow 6x = 21$.

→ $x = 4$. Substitute back to find $y = 3$.

Answer: $x = 4, y = 3$

10. Solve the system: $2x + 1y = 2$ and $2x + 3y = 10$.

$$2x + 1y = 2, 2x + 3y = 10$$

→ From the coefficients and solution, $c = 2(-1) + 1(4) = 2$.

→ $f = 2(-1) + 3(4) = 10$.

→ Using elimination: multiply Eq1 by 2 and Eq2 by 2, subtract to solve for y , then back-substitute.

→ Solution: $x = -1, y = 4$.

Answer: $x = -1, y = 4$

11. Solve the system by substitution: $y = 3x + 4$ and $4x + 2y = -12$.

$$y = 3x + 4, 4x + 2y = -12$$

→ Substitute $y = 3x + 4$ into $4x + 2y = -12$.

→ $4x + 2(3x + 4) = -12 \Rightarrow (4 + 6)x = -12 - 8 \Rightarrow x = -1$.

→ $y = 3(-1) + 4 = -4$.

Answer: $x = -1, y = -4$

12. Solve the system by elimination: $5x - 3y = 19$ and $4x + 3y = 26$.

$$5x - 3y = 19, 4x + 3y = 26$$

→ Add the equations to eliminate y : $(5+4)x = 19 + 26 \Rightarrow 9x = 45$.

→ $x = 5$. Substitute back to find $y = 2$.

Answer: $x = 5, y = 2$

13. Solve the system: $3x + 2y = 0$ and $4x + 2y = -2$.

$$3x + 2y = 0, 4x + 2y = -2$$

→ From the coefficients and solution, $c = 3(-2) + 2(3) = 0$.

→ $f = 4(-2) + 2(3) = -2$.

→ Using elimination: multiply Eq1 by 4 and Eq2 by 3, subtract to solve for y , then back-substitute.

→ Solution: $x = -2, y = 3$.

Answer: $x = -2, y = 3$

14. Solve the system by substitution: $y = 1x + 0$ and $1x + 3y = -6$.

$$y = 1x + 0, 1x + 3y = -6$$

→ Substitute $y = 1x + 0$ into $1x + 3y = -6$.

→ $1x + 3(1x + 0) = -6 \Rightarrow (1 + 3)x = -6 - 0 \Rightarrow x = -3$.

→ $y = 1(-3) + 0 = -1$.

Answer: $x = -3, y = -1$

15. Solve the system by elimination: $5x - 3y = 22$ and $3x + 2y = 17$.

$$5x - 3y = 22, 3x + 2y = 17$$

→ Add the equations to eliminate y : $(5+3)x = 22 + 17 \Rightarrow 8x = 39$.

→ $x = 5$. Substitute back to find $y = 1$.

Answer: $x = 5, y = 1$

16. Solve the system: $1x + 3y = -1$ and $2x + 2y = -6$.

$$1x + 3y = -1, 2x + 2y = -6$$

→ From the coefficients and solution, $c = 1(-4) + 3(1) = -1$.

→ $f = 2(-4) + 2(1) = -6$.

→ Using elimination: multiply Eq1 by 2 and Eq2 by 1, subtract to solve for y , then back-substitute.

→ Solution: $x = -4, y = 1$.

Answer: $x = -4, y = 1$

17. Solve the system by substitution: $y = 1x + -1$ and $1x + 1y = 0$.

$$y = 1x + -1, 1x + 1y = 0$$

→ Substitute $y = 1x + -1$ into $1x + 1y = 0$.

→ $1x + 1(1x + -1) = 0 \Rightarrow (1 + 1)x = 0 - -1 \Rightarrow x = 1$.

→ $y = 1(1) + -1 = -1$.

Answer: $x = 1, y = -1$

18. Solve the system by elimination: $5x - 1y = 9$ and $1x + 2y = 4$.

$$5x - 1y = 9, \quad 1x + 2y = 4$$

→ Add the equations to eliminate y : $(5+1)x = 9 + 4 \Rightarrow 6x = 13$.

→ $x = 2$. Substitute back to find $y = 1$.

Answer: $x = 2, \quad y = 1$

19. Solve the system: $2x + 1y = 0$ and $3x + 2y = 0$.

$$2x + 1y = 0, \quad 3x + 2y = 0$$

→ From the coefficients and solution, $c = 2(0) + 1(0) = 0$.

→ $f = 3(0) + 2(0) = 0$.

→ Using elimination: multiply Eq1 by 3 and Eq2 by 2, subtract to solve for y , then back-substitute.

→ Solution: $x = 0, y = 0$.

Answer: $x = 0, \quad y = 0$

20. Solve the system by substitution: $y = 3x + -1$ and $3x + 2y = -4$.

$$y = 3x + -1, \quad 3x + 2y = -4$$

→ Substitute $y = 3x + -1$ into $3x + 2y = -4$.

→ $3x + 2(3x + -1) = -4 \Rightarrow (3 + 6)x = -4 - -2 \Rightarrow x = 0$.

→ $y = 3(0) + -1 = -1$.

Answer: $x = 0, \quad y = -1$

21. Solve the system by elimination: $2x - 2y = 6$ and $2x + 2y = 14$.

$$2x - 2y = 6, \quad 2x + 2y = 14$$

→ Add the equations to eliminate y : $(2+2)x = 6 + 14 \Rightarrow 4x = 20$.

→ $x = 5$. Substitute back to find $y = 2$.

Answer: $x = 5, \quad y = 2$

22. Solve the system: $3x + 2y = 15$ and $3x + 3y = 18$.

$$3x + 2y = 15, \quad 3x + 3y = 18$$

→ From the coefficients and solution, $c = 3(3) + 2(3) = 15$.

→ $f = 3(3) + 3(3) = 18$.

→ Using elimination: multiply Eq1 by 3 and Eq2 by 3, subtract to solve for y , then back-substitute.

→ Solution: $x = 3, y = 3$.

Answer: $x = 3, \quad y = 3$

23. Solve the system by substitution: $y = 3x + 2$ and $1x + 1y = -3$.

$$y = 3x + 2, \quad 1x + 1y = -3$$

→ Substitute $y = 3x + 2$ into $1x + 1y = -3$.

→ $1x + 1(3x + 2) = -3 \Rightarrow (1 + 3)x = -3 - 2 \Rightarrow x = 0$.

→ $y = 3(0) + 2 = 2$.

Answer: $x = 0, \quad y = 2$

24. Solve the system by elimination: $4x - 2y = 10$ and $2x + 2y = 14$.

$$4x - 2y = 10, \quad 2x + 2y = 14$$

→ Add the equations to eliminate y : $(4+2)x = 10 + 14 \Rightarrow 6x = 24$.

→ $x = 4$. Substitute back to find $y = 3$.

Answer: $x = 4, \quad y = 3$

25. Solve the system: $4x + 2y = -12$ and $2x + 2y = -6$.

$$4x + 2y = -12, \quad 2x + 2y = -6$$

→ From the coefficients and solution, $c = 4(-3) + 2(0) = -12$.

→ $f = 2(-3) + 2(0) = -6$.

→ Using elimination: multiply Eq1 by 2 and Eq2 by 4, subtract to solve for y , then back-substitute.

→ Solution: $x = -3, y = 0$.

Answer: $x = -3, \quad y = 0$

26. Solve the system by substitution: $y = 2x + 3$ and $4x + 2y = -18$.

$$y = 2x + 3, \quad 4x + 2y = -18$$

→ Substitute $y = 2x + 3$ into $4x + 2y = -18$.

→ $4x + 2(2x + 3) = -18 \Rightarrow (4 + 4)x = -18 - 6 \Rightarrow x = -3$.

→ $y = 2(-3) + 3 = -3$.

Answer: $x = -3, \quad y = -3$

27. Solve the system by elimination: $2x - 2y = -2$ and $2x + 2y = 18$.

$$2x - 2y = -2, \quad 2x + 2y = 18$$

→ Add the equations to eliminate y : $(2+2)x = -2 + 18 \Rightarrow 4x = 16$.

→ $x = 4$. Substitute back to find $y = 5$.

Answer: $x = 4, \quad y = 5$

28. Solve the system: $2x + 3y = -17$ and $1x + 2y = -10$.

$$2x + 3y = -17, \quad 1x + 2y = -10$$

→ From the coefficients and solution, $c = 2(-4) + 3(-3) = -17$.

→ $f = 1(-4) + 2(-3) = -10$.

→ Using elimination: multiply Eq1 by 1 and Eq2 by 2, subtract to solve for y , then back-substitute.

→ Solution: $x = -4, y = -3$.

Answer: $x = -4, \quad y = -3$

29. Solve the system by substitution: $y = 3x + 3$ and $4x + 1y = -12$.

$$y = 3x + 3, \quad 4x + 1y = -12$$

→ Substitute $y = 3x + 3$ into $4x + 1y = -12$.

→ $4x + 1(3x + 3) = -12 \Rightarrow (4 + 3)x = -12 - 3 \Rightarrow x = -3$.

→ $y = 3(-3) + 3 = 0$.

Answer: $x = -3, \quad y = 0$

30. Solve the system by elimination: $2x - 1y = 1$ and $3x + 1y = 14$.

$$2x - 1y = 1, \quad 3x + 1y = 14$$

→ Add the equations to eliminate y : $(2+3)x = 1 + 14 \Rightarrow 5x = 15$.

→ $x = 3$. Substitute back to find $y = 5$.

Answer: $x = 3, \quad y = 5$
